

A Conditional Diffusion Framework for Sample-Efficient Thermal Modeling in 3DICs

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Abstract—Thermal constraints pose a critical challenge in modern three-dimensional integrated circuits (3DICs) due to the increasing density of transistors and the limited heat dissipation paths. We propose a conditional diffusion-based generative framework that accurately predicts static temperature distributions from power maps with low latency. Our model introduces a fusion-based U-Net architecture named HeatDiffUNet that integrates power and temperature information across multiple scales, improving conditioning effectiveness and thermal localization. Our proposed method achieves a temperature difference error of 2.31°C using only 200 training samples, outperforming existing methods by as much as 68%. Notably, even when provided with five times more training data, both the prior techniques are unable to match the predictive accuracy demonstrated by our model. Additionally, our approach excels in hotspot localization, where we achieve up to a 37% performance uplift, despite baselines being trained on twice the amount of data. These results underscore the suitability of our method for accurate thermal analysis in early-stage 3DIC design. Our code is publicly available at <https://github.com/FinAminToastCrunch/HeatDiffUNet>

Index Terms—3DIC, diffusion, thermal, sample-efficiency

I. INTRODUCTION AND BACKGROUND

Modern ICs, characterized by high transistor densities and operating frequencies, increasingly face critical thermal management challenges. Ignoring thermal issues risks device failures, such as electromigration and dielectric breakdown, as well as performance losses due to higher interconnect resistivity, resulting in increased IR drops and RC delays. 3DICs, which vertically stack logic and memory layers, mitigate some performance bottlenecks but intensify heat generation and restrict cooling capabilities, complicating thermal management compared to planar architectures [1; 2].

While finite element methods (FEM) remain the gold standard for thermal prediction accuracy [2], their high computational cost and precise physical modeling limit scalability [3]. An early acceleration effort for full-chip thermal analysis—a geometric multigrid solver [4]—improved speed and memory efficiency. It used bin-averaged layer conductivities rather than explicit layout details. Machine learning methods have since emerged as promising alternatives. System-level techniques [5] capture coarse thermal trends, while deep neural networks [6] and physics-informed frameworks [7] enhance resolution and incorporate domain constraints. Autoencoder-based ML solvers [8] handle boundary conditions, further improving

modeling fidelity.

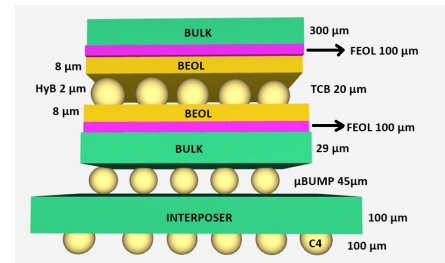


Fig. 1: Cross-sectional view of a 3DIC stack with labeled layers and interconnect dimensions used for data creation [9]

Among recent learning-based approaches, ThermEDGE [10] and the conditional generative adversarial network (cGAN) [11] serve as effective baselines for power map-conditioned thermal prediction. The cGAN deploys a method that replaces the typical input noise vector with dropout during training to simulate randomness in the generator. ThermEDGE adopts a U-Net-style encoder-decoder trained on spatial power-temperature pairs. Both models, however, are not sample-efficient and require large training datasets to achieve reasonable prediction accuracy. This is a critical limitation in early-stage 3D-IC design, where thermal modeling must operate under limited data and runtime constraints. Smith et al. [12] also adopt a cGAN-based approach for thermal prediction, conceptually similar to that of Kashyap et al. [11]. Due to the lack of available implementation details for Smith et al., we adopt Kashyap et al.’s model as our primary cGAN baseline.

So, developing rapid and sample-efficient thermal models that can accurately localize hotspots is crucial for iterative optimization during the early stages of design. A simulator that efficiently handles diverse power patterns within a fixed floor plan greatly enhances early thermal analysis for 3DIC design. Our work investigates a conditional diffusion model for this task.

II. PROPOSED ARCHITECTURE

Conditional Denoising Diffusion Probabilistic Models (cDDPMs) [13] reverse a forward noising process guided by auxiliary inputs. Unlike traditional DDPMs, which predict noise, we directly estimate the clean image x_0 from a noisy

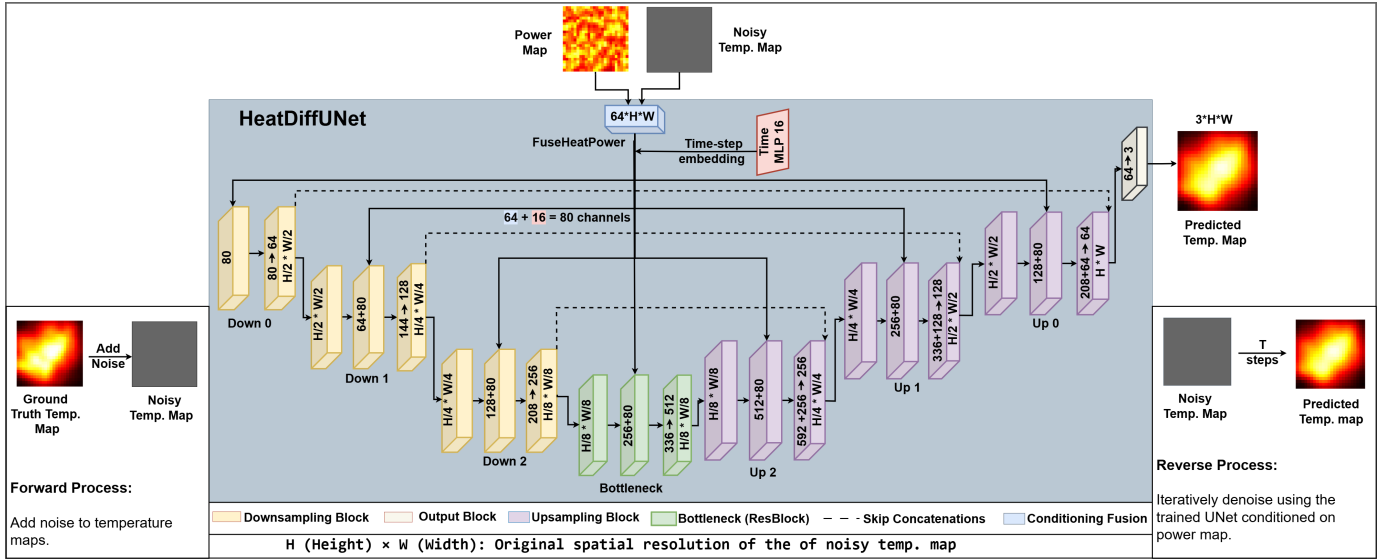


Fig. 2: Proposed cDDPM architecture (HeatDiffUNet). Power and noisy-temperature maps are fused; at each resolution, a small MLP-derived timestep embedding is broadcast over the spatial grid and concatenated with the fused features before each encoder/decoder block of the multi-scale U-Net with skip connections.

input $\mathbf{x}_t$. The forward process progressively adds Gaussian noise over T timesteps:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}) \quad (1)$$

The marginal distribution is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (2)$$

with $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$.

The reverse (generative) process is modeled by:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t, \mathbf{c}), \sigma_\theta^2 \mathbf{I}) \quad (3)$$

Here, μ_θ is a neural network parameterized by θ , and $\mathbf{c}$ is the conditioning input (e.g., a power map).

The training objective uses a composite loss

$$\mathcal{L}_{\text{denoise}} = \lambda_1 \|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|_2^2 + \lambda_2 \text{SoftDice}(\mathbf{x}_0, \hat{\mathbf{x}}_0) + \lambda_3 D_{\text{TV}}(\hat{\mathbf{x}}_0) \quad (4)$$

Here, $\hat{\mathbf{x}}_0$ is the predicted temperature map. The loss combines three terms: mean squared error for pixel-wise accuracy, soft dice loss for structural alignment, and total-variance loss for spatial smoothness. The sampling starts from Gaussian noise and iteratively refines the sample using the learned reverse process. This predict-clean formulation is an alternative parameterization in DDPM [14].

As illustrated in Figure 2, we propose a cDDPM designed to predict temperature distributions in 3DICs from given power maps. We designed our model upon a customized U-Net architecture [15] named **HeatDiffUNet**, which incorporates multi-resolution conditioning [16] through learned feature fusion [17], and employs timestep embeddings mapped by a small MLP [14; 18]. Specifically, the network receives a noisy temperature map, the associated power map, and a normalized timestep t , aiming to generate a clean temperature map at each denoising step. Fusion is implemented via a dedicated module **FuseHeatPower**, which initially processes the temperature and power maps independently through convolutional layers,

subsequently concatenates them, and merges them with a 1×1 convolution (Figure 3).

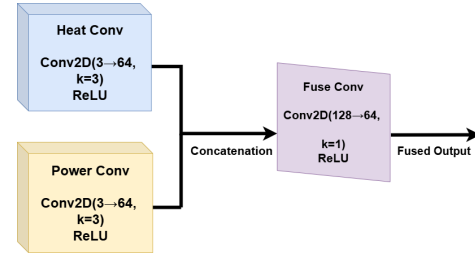


Fig. 3: Proposed FuseHeatPower module architecture for merging power and temperature map features.

This fusion strategy follows standard feature-level fusion practice [17], specifically adapted here to combine power-temperature inputs. Concurrently, a small MLP transforms the normalized timestep into a channel embedding [18]. This embedding is broadcast across the spatial grid and concatenated with the fused power-temperature features at each encoder and decoder stage. Our complete model architecture comprises three downsampling blocks, a bottleneck residual block, and three corresponding upsampling blocks, connected by skip concatenations to retain fine-grained spatial information. A final 1×1 convolution outputs the predicted clean temperature map. During training, Gaussian noise is incrementally introduced to the ground truth temperature maps over T timesteps, following a linear schedule. The model optimizes a composite loss (Equation 4) that directly predicts the clean map ($\mathbf{x}_0$) from the noisy input ($\mathbf{x}_t$). The foundational diffusion process follows the DDPM framework by [14], which leverages layout-specific power data. At inference, the process begins from pure noise, progressively denoising to yield precise thermal predictions.

III. EXPERIMENTAL SETUP AND DATASET CREATION

We obtained our dataset following the same methodology proposed by Kashyap et al. [11], utilizing the same 3D stack structure (shown in Figure 1) and identical boundary conditions.

TABLE I: Temperature difference compared to ground truth ($^{\circ}\text{C}$) across training set sizes (lower is better)

Model	200	300	400	500	600	700	800	900	1000
Our cDDPM	2.31	1.69	1.43	1.21	1.17	1.06	1.03	0.99	0.95
cGAN [11]	4.10	3.56	3.21	3.09	2.97	2.81	2.68	2.54	2.40
ThermEDGE [10]	7.26	7.19	7.15	7.16	7.13	7.09	7.12	7.09	7.07

These include heat transfer coefficients of $107 \text{ W/mm}^2\text{K}$ at the top, and $33 \text{ W/mm}^2\text{K}$ and $200 \text{ W/mm}^2\text{K}$ for the sides and bottom, respectively. Siemens EDA[®] Calibre 3DThermal software was used to generate ground truth temperature maps from randomized power distributions. The resulting dataset consists of power and heat maps at a 19×19 and 25×25 resolution, respectively.

We generated 1998 paired power-temperature maps. From this set, we allocated 100 for validation, 200 for testing, and used the remaining data to vary the training set across six pre-defined sizes systematically. This experimental setup allowed us to evaluate not only the absolute predictive performance but also the sample efficiency, a crucial consideration for early-stage IC design scenarios where labeled thermal data is often limited. For each of the specified training set sizes, we benchmarked our proposed cDDPM model against two established baseline models: cGAN [11] and ThermEDGE [10]. Predictive accuracy and hotspot deviation were measured and reported on the fixed 200-sample test set. The hotspot deviation was computed by locating the maximum temperature in the predicted map and comparing its position with the ground truth maximum temperature location.

IV. RESULTS AND DISCUSSIONS

One of the most striking observations from our experimental results in Table I is the exceptional performance of the cDDPM, even under minimal data availability. When trained with just 200 samples, cDDPM achieves a temperature difference of 2.31°C compared to ground truth—a level of accuracy unmatched by cGAN or ThermEDGE, even at fivefold higher training sizes. Specifically, cDDPM is approximately 3.75% more accurate than cGAN and 67.33% more accurate than ThermEDGE at their maximum training sizes. This advantage can be attributed to the cDDPM’s multi-scale fusion of heat and power data, and its iterative denoising process, which inherently models uncertainty and enables superior generalization on unseen data and sample efficiency.

TABLE II: Average of max. temperature location deviation (pixels) across training set sizes (lower is better)

Model	200	300	400	500	600	700	800	900	1000
Our cDDPM	4.15	3.08	2.97	2.47	2.52	2.33	2.28	2.16	2.08
cGAN [11]	4.86	4.62	4.56	4.39	4.35	4.25	4.16	4.00	3.93
ThermEDGE [10]	3.24	3.09	2.85	2.82	2.81	2.74	2.64	2.55	2.50

Table II illustrates hotspot localization performance. At 500 samples, cDDPM’s average deviation is 2.47 pixels, notably

better than both baselines. Increasing the sample size to 1000 further reduces this deviation, while cGAN and ThermEDGE remain largely unchanged.

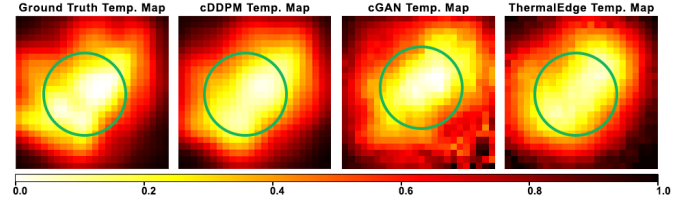


Fig. 4: Comparison of ground truth and predicted maps with hotspot regions highlighted. Hotspots are enclosed in circles to help visualization.

This suggests that our cDDPM is not only learning temperature distributions but also internalizing the underlying spatial structure. Thus, enabling more precise identification of critical thermal regions with limited supervision.

V. CONCLUSION

Overall the experiments demonstrate the superior **sample efficiency** of the cDDPM. A model that can generalize accurately after being trained on a small dataset not only reduces data preparation overhead but also accelerates the thermal analysis loop. These results position cDDPM as a highly effective and resource-efficient solution for predictive thermal modeling in practical design scenarios in early stages of 3DIC design.

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